

Investigating the Effects of Different Levels of User Control on the Effectiveness of Context-Aware Recommender Systems for Web-Based Search

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ABSTRACT

Recommender systems assist users by providing recommendations based on some filtration criteria to reduce information overload. Embedding context-awareness allows recommender systems to use context information around the user, situation, and system to adapt and provide more efficient, relevant, and personalized recommendations. However, embedding context-awareness into recommender systems inherently limits the users' control over the systems due to reduced interactivity from automatic adaptations. This may potentially impact users' use and perception of the systems. Control can be purposefully designed to be given to the user in context-aware recommender systems at different levels. Our work investigates the effects of different levels of user control on the effectiveness and understandability of context-aware recommender systems (CARS) within the scenario of learning through web-based search (called 'Search-As-Learning'). To enable our study, we implemented a CARS that supports web-based search by recommending users a link using context such as browsing history. Our study found that participants used more recommendations from the CARS with high control compared to no control and some control. In conclusion, higher control in a recommender system for web-based search is preferred by the user despite control manipulation taking more time possibly due to explicit user needs.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; *User studies*.

KEYWORDS

User Control, Digital Context, Context-Awareness, Online Learning, Search as Learning

ACM Reference Format:

Neha Rani, Sharon Lynn Chu, and Victoria Rene Mei. 2022. Investigating the Effects of Different Levels of User Control on the Effectiveness of Context-Aware Recommender Systems for Web-Based Search. In *New Orleans '22: ACM Symposium on Neural Gaze Detection, April 29–May 05, 2022, New Orleans, LA, USA*. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3491101.3519802>

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New Orleans '22, April 29–May 05, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519802>

1 INTRODUCTION

Recommender systems have become a valuable tool to provide recommendations to the user, in areas that otherwise are flooded with a surplus of information or resources or items. Recommender systems are commonly used in areas such as online shopping, movie recommendation, music recommendation, tourism, etc. [2, 13, 21]. With the rising popularity of recommender systems, it is now being extended to areas, such as online learning [2, 6, 16, 21]. Often people use web browsing to learn about things and find the answer to their queries in day-to-day life. When browsing online for information, users are often presented with thousands of search results. Often, users end up spending time choosing the resource, which would have been well spent on reading the web page otherwise. This situation points towards the need for a system that supports web-based search and reduces information overload. Recommender systems have been found to help users in surplus of information and resources during web searches [4], and this research area is generally called 'Search-As-Learning' [10]. Search as Learning pertains to search session performed with learning intent which "differ from transactional or navigational search due to dedicated learning intent" [10].

Currently, recommender systems use most commonly some form of a model to predict user preference by observing and analyzing user behavior over time or using similar-user ratings. While these systems use a generalized model to predict user preferences, it might not always result in the most relevant recommendation for the user as it misses the users' current state, current preferences, browsing context, and context of use. Embedding context-awareness in recommender systems will allow it to use contextual information to provide more relevant and accurate recommendations [2]. Using a context-aware recommender system for Search-As-Learning would support the contextualization need and filtering of search results for a web-based search.

However, while context-awareness helps in making automatic adaptation and providing more relevant results by utilizing user information, it takes away control from the user [3, 7]. Barkhuus and Dey [3], in their study, found that the user feels less in control when using either active or passive context-aware system as opposed when personalizing their own applications. Further, context-awareness brings other issues, such as whether the context information is correct, what context information is being used, how the system adapts, etc., where user control can help. Even when using a recommendation system without context-awareness, user control is important as there is a possibility that the information that goes into decision-making for the recommendation is not relevant to the

user or is incorrect, or outdated, or user preferences have changed, or the context of use is different [11].

Providing user control through interactive explanation can help the user by making them aware of the information being used by the system and at the same time allowing them to alter the information. He et al. [9] also mention in their survey of interactive recommendation, integrating user control as an essential criterion. In the context of a recommender system, user control is defined as system manipulation (of context or context information) by the user to affect systems output (recommendation) [11]. Jannach et al. [11] also pointed out that giving some control to the user to specify their preference and able to see and control the information that goes into deciding the recommended item helps overcome the challenge of incorrect information and irrelevant recommendation. Further, increased interactivity with the system increases user satisfaction, and trust [17]. Jannach et al. [11], in their work on user control in a recommender system, pointed out the need to explore the level of control that users in different settings most prefer to identify needs and effects of user control. Giving users the power to control the system's context will allow them to fetch better results with higher efficiency. Further, not having control can also create a sense of anxiety [7]. The user control can serve as a source of feedback, allowing the recommendation system to understand user preferences and further learn and improve.

This paper presents work that explores the conundrum of giving control and making an automatic context-aware recommendation by investigating the effects of offering three different levels of user control in CARS to support learning through web searches. We investigate three levels of controls, namely no control, some level of control, and a high level of control when using CARS for web-based search. Through the user control design in our system, we also show users can control at a different level the context information that goes into CARS. While the concept of controlling context information was proposed earlier, we show a scenario of controlling context in a web-based search for learning. This work contributes to understanding the concept of user control in CARS and the effect of different levels of user control on users' perceptions. We explore differences in user perception of system effectiveness. We also discuss implications on other factors such as user preference.

2 BACKGROUND

Recommender systems are a leading tool that accelerates the information-seeking process by providing a valuable and relevant recommendation. Recommender systems commonly use information, such as user ratings, also called collaborative filtering or similar user preferences, user profiles, etc. Collaborative filtering technique involves using multiple user ratings, then based on these ratings, suggest highly-rated items for similar other users [22]. However, these initial recommender systems lacked the utilization of the user's contextual information to further improve recommendations [8]. Using context-awareness is a recent trend in recommender systems that allows leveraging the contextual information around the user in addition to the usual two-dimensional (user and item) approach for recommendation [1, 8].

While commonly, recommender systems indirectly use user information to generate recommendations, a few directly input user

information through user interaction. Through control, the user can directly control the CARS by altering the information used as the basis of suggestion. Integrating user control into recommendation systems is gaining importance as it allows users to interact with the recommender system and navigate recommendations to their requirements and preferences [9, 20]. This is a transition of recommender system from using others' preferences (i.e., collaborative filtering) to using users' preferences to recommend them. Prior work has explored controlling the recommender system through various ways, such as controlling system settings and attributes, through user feedback on recommendations, sorting or filtering attributes and values [15], providing weight on attributes used in recommendations [12], slider control on system attributes [24], or altering parameters used in recommendation algorithm, etc. [12, 22].

3 RELATED WORK

A user control can allow the user to navigate the process of information retrieval and filtering. Controlling supports interactivity by taking user input and making adaptations accordingly. Prior literature has emphasized the importance of user control in incorporating interactivity and transparency in recommender systems [9]. Prior work in recommender systems has also shown a positive relationship between user control and satisfaction [19]. Through control users can steer recommendations system in the direction they want, which increase perceived quality of the system [18], satisfaction with the system [19] and even increases trust [5], all which can contribute towards higher system acceptance. User control can also increase the user's understanding of the system by observing the input and the corresponding output and gives the user a chance to change the input to get improved recommendations [23]. Parra and Brusilovsky [19] found a positive effect on perceived understanding through controllable interactive recommendation. Jin et al. [13] explored different levels of user control in the music recommender system for contextual characteristics and music preferences. They found that allowing users to control the context can increase users' perceived quality of the system. They also found that scenarios of use will affect the level of control preferred by the user. Further, researchers have pointed out that the one-size-fits-all approach does not necessarily apply to recommender systems in terms of user control [14]. Jin et al. [12] investigated three levels of control and found that most people preferred low or medium levels of control. However, those who are experts on the recommender system preferred high-level control. They also found that a high level of control requires a higher cognitive load and generated high-quality recommendations. Knijnenburg et al. [15], in their investigation of interaction with recommender system, found that novice users prefer a system that directly recommends highly-rated items. In contrast, experts prefer a hybrid recommendation system with implicit and explicit user involvement in preference selection. Personality parameters related to the purpose of the recommender system are also likely to affect user control. For example, in a music recommendation system, users with music sophistication are better able to leverage user control to explore recommendations [14]. The literature is rich in the exploration of user control in different types of recommendation systems. Still, it lacks an understanding of

user control in learning-based or web-search-based recommender systems.

4 STUDY DESCRIPTION

This study was conducted and moderated remotely using teleconferencing software. A within-subjects study design was used for the study. Study conditions were counterbalanced. The level of user control was used as an independent variable. There were three study conditions with variations in the level of user control, namely *no user control*, *some user control*, *high user control*. A chrome plugin-based digitally context-aware learning assistant was developed and used for the study, which is described in further sections.

4.1 Participants

A total of 31 participants took part in the study. The university's ethics board approved the study. Participants were recruited from the university undergraduate pool. Signed consent forms were obtained before the study, and the participants were free to opt-out of the study without any consequences. Participants took approximately an hour and 10 minutes to complete the study. They were compensated with course credit for participating and completing the study. The average age of the participants was 20.32 years. The study population consisted of 21 male and 10 female participants. There were 17 White/Caucasian, 12 Asian, 1 Black/African American, and 1 Other ethnicity participants.

4.2 System Description

The system used in the investigation of user control consisted of a chrome plugin. This chrome plugin was named 'Learn Better' and is a recommender system for web-search-based learning. The plugin suggests pages to learn, using digital context-awareness. Digital context is defined as the context originating from browsing information gathered by the system. Context information includes the content type (text/video) of digital information displayed, time spent on browsing a page, site domain (for example .edu), the topic of browsed pages. By providing topics not covered and sites not visited previously, our recommender system makes the recommendations to be relevant and novel (new, not visited before). The content type and site domain serve as a medium to provide a diverse format of information delivered online. A local database was developed covering major topics within the topics of debate that were used in the study. This database consisted of web page links covering major topics within the area, different content types (i.e., both video and text versions of information on the same topic), and links of major site domains (.gov, .edu, .org, and .com) with information on the topic. There were four parts within each debate topic: introduction to the topic, its causes, effects, and solutions, with variations depending on the topic. The database had web page links with a focus on these sub-topics. The researcher chose these web pages to make sure they had the content needed, and these pages were usually found on the first page of google search. Based on the context information, the topic entered, the system traverses the decision tree and a web link from this database was recommended. The decision tree used by the system is shown in Figure 1.

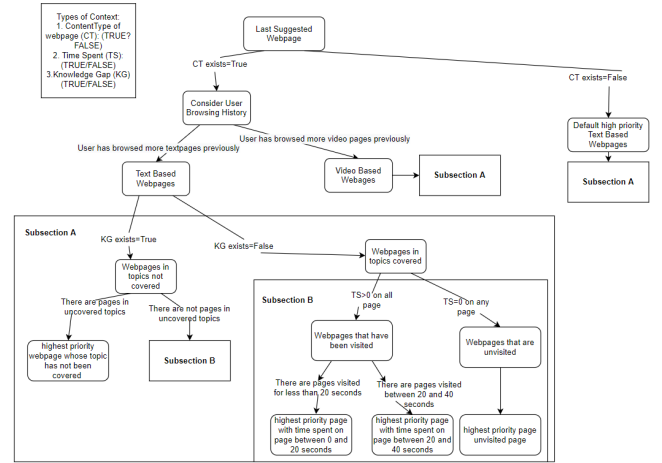


Figure 1: The decision tree used by the CARS with different levels of control to recommend web pages based on entered topic and the context information.

There were three versions of the plugin based on control conditions. One was the system with *no user control*, the other had *some level of user control*, and the other had *higher level of user control*. In the context of our study, control is the ability to control system parameters (context information) to alter system output. We define some control as giving users limited control such that they can control parameters, but on a broader level, they can control the type of context information that goes into making the recommendation. High control means a system with control parameters such that the user can fine-tune parameters such that the user could control the type and value of context information that goes into making the recommendation. All three versions of the learn better application is shown in Figure 2. In all three versions of the interface, users were required to enter the topic they wanted to learn.. In the system with no user control, a page was suggested to the user based on the topic entered and digital context-awareness. In the system with some user control, after entering the topic, the user had the option to control the system to some level by selecting and deselecting the digital context used in suggesting the page. In a system with high user control, the user could control the system by selecting and deselecting options. This time user could control the system at a more keener level and control what context is being selected among the broader digital context used.

4.3 Study Procedure

Main study task was to use the plugin to learn about a topic and then debate on the topic based what was learned. Debate was used as a motivation for learning, test or MCQ was not used as it would be more of remembrance measure. The learning topics used by were therefore a debate topic where there are both pro and cons. The study was conducted in pairs, meaning two participants took part in each study session simultaneously. The study task was to learn about a given topic using the 'Learn Better' plugin and prepare for a debate. Both participants prepared for the debate on the given topic by using the same version of the plugin. Participants formed

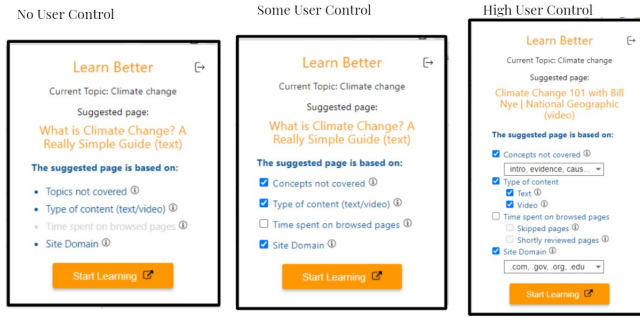


Figure 2: Learn Better Application: this a recommender system for search as learning. All three versions of the application are shown. Left - no user control, mid - some user control, right - no user control

their stance based on what they learned and then participated in a debate with one another.

The study started with participants' and researchers' introduction. Then the participants were briefed about the study and the study task. Then the debate topic was assigned to both the participants, and they used the same version of the plugin to prepare for the debate. Two breakout rooms were created to have separate sessions for each participant to learn independently using the plugin. There were two researchers to moderate the study. A researcher accompanied each participant. Researchers shared their screen, where the plugin was installed. Participants controlled the researcher's screen to use the plugin and learn about the debate topic. The plugin was installed on the researcher's computer and not the participants' computer to avoid data privacy issues. Participants were explained how to use the plugin and were given 2 minutes to try out the plugin themselves. Participants had 8 minutes to use the plugin and prepare for the debate on the given topic. Participants were free to use the plugin as many times as they wanted to get recommendation of links to read from. After participants prepped for the debate, both breakout rooms were merged, and the debate process started. Each participant was asked to present their stand on the topic and their opinion. This entire process was repeated three times for each study condition as it was a within subjects study. A new debate topic was assigned for each condition, and a different version of the plugin was used. Post-study questionnaires were given to both the participants after each debate. The study ended with a semi-structured interview.

The research question we aimed to investigate through our research was: **Does different levels of user control affect the effectiveness and perception of a context-aware recommender system for Search-As-Learning?**

5 MEASURES AND DATA ANALYSIS

Following measures were evaluated in the study. The first measure was **effectiveness** of the system, which is a measure of how well the system recommends such that users were able to get web page or information they desired. Another measure evaluated was **efficiency** which tells how easy to use, and productive the system was in getting the desired recommendations. **Understanding of**

the system functioning was evaluated. **Preference of the system** with different levels of control was also evaluated. Parametric tests were used for statistical analysis if the data was normal or else non-parametric tests were run. For all the statistical tests, 0.05 was the threshold for significance.

Effectiveness of system was measured using the number of recommended web-pages that were visited/read by the participant during the study task. Count of recommended pages opened by the users was listed on a spreadsheet. The effect of user control on overall **efficiency** of system in supporting learning was evaluated through two interview question: (i) "Which one of the three made you more efficient?" and (ii) Does having control over the application help increase your efficiency of learning?

Two measures were used to analyze participants' **understanding of the system**. These were a perceived understanding questionnaire and interview questions. A single-item questionnaire on a 7-point Likert scale was used. The question used was "I understand the working of the plugin fully". ANOVA with repeated measures test was performed on the participants reported score. The response to the interview question "Does controlling the system help increase your understanding of the system? your choices are yes, no and maybe" was evaluated. Count of yes, no, and maybe in the response is reported. To find participants' **preference of the system**, they were asked the interview question "Which of the three systems would you use in your daily life?". A Chi-square test was run to find any significant relationship between the level of control in the system and users' preference of the system.

6 RESULTS AND DISCUSSION

Results of the Friedman test showed a statistically significant difference in the number of suggested pages visited by the participant depending on user-control level, $\chi^2(2) = 6.090$, $p = 0.048$. The number of suggested pages opened was highest for high control (Mean=4.06), followed by no control (Mean=3.48) and some control (Mean=3.42). This indicates that participants found the system with high control to be more useful and *effective*; therefore they used more suggested pages compared to no control and some control. Almost no difference in the number of suggested pages opened between some and some control can be attributed to the possibility that little control does not give enough freedom to the users get the page they wanted. When the user knows what they want, they prefer to control, which is the case in explicit web-based search, as there is an objective behind a search that encourages the user to do that exact search. A somewhat similar observation has been in the prior work. Tsai and Brusilovsky [24] found a controllable social recommender system to be rated higher in terms of perceived effectiveness compared to the one without control (evaluated through self-reported measures). Higher control gives users the ability to control the recommender system and its recommendations to get what they want. Adding control to the recommender system adds valuable information to the recommender system on top of users rating, preferences, and context around the user. Please note that such user control differs from user preferences as preference is what users generally like, whereas control accommodates different requirements in each instance of using the recommender system.

In response to an interview question, 30 out of 32 participants mentioned that controlling the system made them more *efficient*. Multiple participants mentioned that they were better able to manipulate system control with a high control system to get what they wanted. They also mentioned spending more time on a system with higher control as they interacted with more controls and used more recommended pages. A couple of participants also mentioned that sometimes they felt that controlling the system meant putting more time, which is not surprising. In web-based search, where user needs can be diverse, control can serve well, but there are other considerations to make, such as, the system should provide user control such that it's simple to use and less time-consuming for it to remain valid. We used a simple selection mechanism as a form of control for this study. We did not evaluate participants' performance during the debate as a measure of learning because debate performance can also depend on a number of factors, such as speaking skills and prior knowledge. However, this is not particularly a limitation as learning is likely to differ based on the web page content when using different web pages and is not directly dependent on the control level.

No significant difference was observed on the average score of a self-reported measure of *understanding*. However, the interview question response showed that the majority of the participants (29 out of 31) agreed that controlling the system helps increase understanding of the system. This could be potentially due to user control supporting increased interaction leading to the perception of supporting understanding. We can say that although there is no significant increase in understanding, there is a general positive inclination towards control supporting users' system understanding. Prior literature has similar finding in for user control in other interactive recommender system such as controllable social recommender system [9, 24].

The chi-square test results show a significant association between level of control and users' *preference* for recommender system $\chi^2(2) = 43.969$; $p < .001$. High control was preferred 78.1% of the times, some control 15.6%, and no control 6.3% of the times. This preference for higher control rather than some control is contradictory to the finding of Jin et al. [13] where they found that some control is preferred by most users and the high control is preferred only by experts. A possible explanation is that web-based search is commonly used by people, making them almost an expert, leading to the preference for higher control. In other recommender systems such as e-shopping, music recommendation systems, users get recommendations while using the application without explicitly asking, which can make lower control preferred. This preference for a system with higher control is potentially due to the ability to fine-tune the system to get what the user wants. However, few participants who preferred a system with no or some control said they would use it when they just wanted basic information on the topic and were not looking for something specific. Context of use can be one important factor to explore in future research.

7 CONCLUSION AND FUTURE WORK

We investigated different levels of user control in a context-aware recommender system for web-search-based learning. We used on-line context such as users browsing history. Our investigation found

that when using a recommender system for common everyday tasks such as web-based search, high control is preferred by the users due to higher effectiveness. Although not with concrete evidence, we found that incorporating control supports understanding of the system. Overall, when designing a recommender system for a everyday activity such as a web-based search, higher user control may support higher acceptance. However, the control design should be simple and easy for users to use effectively. Our work lays the ground for developing an understanding for the designers of context-aware recommender systems.

Future work should explore context-aware recommender systems using other types of context and other forms of control to validate our findings and investigate applicability in different closely related contexts. While we investigate controls in the form of a selection of options, using other forms of control which are more complex can lead to different user preferences. Further exploring differences in learning outcomes when using different levels of control in CARS for web-based search for learning outcome needs to be explored. For future work, we would like to explore the difference in motivation and interest in learning when the system automatically recommends learning resources as compared to a user actively involved in the search process.

8 LIMITATIONS

We acknowledge that the algorithm used by our recommender system was rather simple. Still, we believe this will not affect the study results as we did not show or discuss the algorithm participants. Also, as we were interested in exploring the effect of different levels of control, developing a fully comprehensive recommender system was not the priority.

ACKNOWLEDGMENTS

We acknowledge and thank Yadi Qian for her help in initial design of the interface used in the study. We thank all the participants who took part in the study.

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